FACULTY OF INFORMATION TECHNOLOGY

IT PROJECT 700 | GROUP 55

EMERGEX

Centralized Student Accommodation & Secure Booking Application

PHASE 1 - PROJECT PROPOSAL

Purpose, implementation language and SDLC model

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Project lead baseline: Tristan Moses | Every member contributes and presents.

Project Proposal

1. Purpose of the Project

EMERGEX is a centralized South African student accommodation search and secure booking system. It is a response to the Department of Higher Education and Training's constant accommodation shortage report and the practical challenges of comparing trustworthy rooms, timely landlord decisions and deposit risk management. Students can search for verified properties through location, institution, price, whether NSFAS status and Instant Book. You will see rent, deposit, rooms, amenities and visual-evidence categories such as walkthroughs, floor plans and 360-degree tours.

The booking workflow will display every state. You can initiate an Instant Booking or send a new travelling request. The simulated escrow record will not-paid and held and released states. The release will only be after student check-in is confirmed. Funds that exist will proceed through. Administrators will validate landlords, properties, and uploaded demo evidence and audit events, while landlords will handle their own requests. The goal is to provide a clear, secure and sensitive alternative to fragmentation discovery, casual status monitoring and direct deposit handling.

2. Implementation Language

The LAMP stack will be used in a Windows XAMPP environment application. Using HTML5 and CSS3 for mobile-first simplicity, javascript will progressively enhance the navigation and form behaviour. The framework comprises of the performance attributes needed in PHP 8 such as data caching and caching strategies to accelerate performance. MySQL/InnoDB organizes users, properties, bookings and so on with FK, constraints, indices and transactions. State will be escrowed. Documents and audit logs will also be stored, naturally. The core security baseline will consist of PDO prepared statements, password hash/password_verify, CSRF tokens, escaped output and server-side role checks.

3. SDLC Model

The software development lifecycle will utilize Waterfall-agile SDLC. The sequential academic deliverables are proposal, planning, analysis, system design and implementation. The evidence produced in each gate is subject to supervisor review before the next baseline is finalized. The cross-functional agile team will consist of around six people working from a dedicated gate in agile sprints. We will conduct weekly integration reviews and use a traceability matrix for monitoring requirements.

Table 1. Phase 1 design summary

Design decision	Approved direction
Primary users	Students, landlords and platform administrators.
Core value	Verified visual listings, responsive booking, transparent states and simulated deposit protection.
Technology	HTML5, CSS3, JavaScript, PHP 8, Apache and MySQL/InnoDB via XAMPP.
Delivery model	Waterfall phase approvals with Agile implementation sprints.
Safety boundary	Demo data and simulated escrow only; production payment/privacy compliance is future work.

Selected sources: DHET student housing review and symposium; NSFAS accommodation guidance; POPIA Act 4 of 2013; PHP and OWASP security guidance. Full URLs appear in the integrated final report.

4. Project Budget & Resource Cost Estimation

To test financial viability, the budget for the development of the EMERGEX prototype has been calculated using standard industry software engineering rates for a 6 person development team, presumably students, for the duration of a simulated project. The estimated total baseline cost of R128,799.00 takes into account development hours, the cost of a LAMP stack (Zero cost for Apache, PHP and MySQL as they are open source), test rig etc.

Simulated Budget for EMERGEX (Development Phase)

Cost Estimation Technique Used: Bottom-Up Estimation (breaking down individual components to find the total).

Category	Description / Justification	Estimated Cost (ZAR)
1. Human Resources (Labor)	6 Junior Developers (EMERGEX) working part-time. (Estimated: 720 total hours @ R150/hr)	R 108,000.00
2. Cloud & Hosting Infrastructure	Web Server Hosting (e.g., Afrihost or AWS EC2) for backend PHP deployment & database hosting for 6 months.	R 2,500.00
3.Domain Registration	Registration of the emergex.co.za domain name for 1 year.	R 99.00
4. Third-Party APIs	Google Maps API (for property location search) and Twilio SMS API (for Instant Booking notifications).	R 1,500.00
5. Payment Gateway Setup	PayFast / Yoco merchant account setup (simulating the Escrow deposit system).	R 0.00 (Transaction fee based)
6. Software & Development Tools	UI/UX Design (Figma), Code Repositories (GitHub), and Testing Frameworks (bug-test-scripts).	R 0.00 (Using Student/Open Source)
7. Marketing & Launch Simulation	Simulated budget for social media ads targeting university students for the platform launch.	R 5,000.00
8. Contingency Buffer (10%)	A standard 10% buffer for unexpected technical roadblocks or server upgrades.	R 11,700.00
TOTAL ESTIMATED PROJECT COST	**Calculated Capital Required to Build and Launch EMERGEX**	**R 128,799.00**

Sources such as the DHET student housing review and symposium, NSFAS accommodation guidance, POPIA Act 4 of 2013, and PHP and OWASP were selected. Each URL appears in the integrated final report.